

Social Planner's Problem on Slide 20

The social planner's problem is to choose an allocation to maximize the sum of utilities subject to the consumption set, technology and feasibility constraints:

$$\max_{\substack{(m_1, \dots, m_I) \\ (x_1, \dots, x_I) \\ (z_1, \dots, z_J) \\ (q_1, \dots, q_J)}} \sum_{i=1}^I [\phi_i(x_i) + m_i]$$

subject to:

1. $x_i \geq 0$ for all i (consumption set constraint for good ℓ).
2. $q_j \geq 0$ for all j (firm output must be nonnegative).
3. $z_j \geq c_j(q_j)$ for all j (technology constraint for each firm).
4. $\sum_{i=1}^I m_i \leq \bar{\omega}_m - \sum_{j=1}^J z_j$ (feasibility for the composite commodity).
5. $\sum_{i=1}^I x_i \leq \sum_{j=1}^J q_j$ (feasibility for good ℓ).

Because utility is always increasing in both goods, constraints 3-5 will hold with equality. We can therefore write:

$$\sum_{i=1}^I m_i = \bar{\omega}_m - \sum_{j=1}^J z_j = \bar{\omega}_m - \sum_{j=1}^J c_j(q_j)$$

The social planner's problem can then be rewritten as:

$$\max_{\substack{(x_1, \dots, x_I) \\ (q_1, \dots, q_J)}} \sum_{i=1}^I \phi_i(x_i) + \bar{\omega}_m - \sum_{j=1}^J c_j(q_j)$$

subject to:

1. $x_i \geq 0$ for all i (consumption set constraint for good ℓ).
2. $q_j \geq 0$ for all j (firm output must be nonnegative).
3. $\sum_{i=1}^I x_i = \sum_{j=1}^J q_j$ (feasibility for good ℓ).